UNIVERSITI TEKNOLOGI MARA
FINAL EXAMINATION

COURSE	PROGRAMMING I
COURSE CODE	CSC402
EXAMINATION	NOVEMBER 2026
TIME	3 HOURS

INSTRUCTIONS TO CANDIDATES

1. This question paper consists of the questions printed in this booklet.
2. Answer ALL questions in the Answer Booklet. Start each answer on a new page.
3. Do not bring any material into the examination room unless permission is given by the invigilator.
4. Please check to make sure that this examination pack consists of:
   i) the Question Paper
   ii) an Answer Booklet - provided by the Faculty
5. Answer ALL questions in English.

DO NOT TURN THIS PAGE UNTIL YOU ARE TOLD TO DO SO

This examination paper consists of 5 printed pages.

PART A (10 MARKS)

Answer ALL questions.

1. What does the % operator return?
   A. The absolute value
   B. The remainder of integer division
   C. The quotient
   D. The square root

2. Which tool represents an algorithm using standard graphical symbols?
   A. Pseudocode
   B. Flowchart
   C. Compiler
   D. Debugger

3. Which identifier is valid in C++?
   A. total score
   B. total_score
   C. 2total
   D. int

4. Which data type stores a whole number?
   A. string
   B. float
   C. char
   D. int

5. Which part of a for loop is executed after each iteration?
   A. The loop body twice
   B. The update expression
   C. The condition only
   D. The initialisation

6. Which structure chooses between two alternative paths?
   A. for
   B. array
   C. struct
   D. if...else

7. Which statement is most suitable for selecting among several constant cases?
   A. do...while
   B. switch
   C. while
   D. for

8. What does endl do?
   A. Declares an integer
   B. Writes a newline and flushes the stream
   C. Ends the program
   D. Reads a line

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9. Which loop is commonly used to process every array element?
   A. for
   B. break
   C. if
   D. switch

10. Which item is NOT a programming paradigm?
   A. Object-oriented programming
   B. Debugging
   C. Procedural programming
   D. Structured programming

PART B (60 MARKS)

QUESTION 1

This question tests high-level languages and debugging.

a) Define syntax error and logic error.

(2 marks)

b) Explain TWO advantages of high-level languages over machine language.

(3 marks)

QUESTION 2

A program reads three integer values and displays the largest value.

a) Draw a flowchart to find the largest of three values.

(6 marks)

b) Write pseudocode and a complete C++ program for the flowchart.

(7 marks)

QUESTION 3

This question tests operator precedence and assignment.

a) Evaluate the following expressions: (i) 7 + 3 * 2, (ii) (7 + 3) * 2, (iii) 17 % 5, (iv) 17 / 5, (v) 17.0 / 5.

(5 marks)

b) Write C++ statements to declare two integers a and b, read their values, and display their sum, difference, product and remainder.

(6 marks)

QUESTION 4

Study the following array program segment:

```
int a[5] = {2, 4, 6, 8, 10};
int s = 0;
for (int i = 0; i < 5; i++) {
   s += a[i];
}
cout << s;
```

a) Trace the program and state the output.

(10 marks)

b) Rewrite the segment using a do...while loop that sums only values greater than 5.

(5 marks)

QUESTION 5

A shop stores the prices of five items in an array.

a) Declare an array named prices with the values 20.5, 55.0, 15.9, 75.0 and 10.0. Display all values.

(8 marks)

b) Write a program segment that applies a 10% discount to prices above RM50 and stores the results in a second array. Display both arrays.

(8 marks)

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PART C (30 MARKS)

QUESTION 1

A bookstore records up to five books. Each book has a title, price and quantity in stock.

Define a struct Book. Write a complete C++ program that reads the records, calculates the total inventory value, and displays the title of the most expensive book.

(15 marks)

QUESTION 2

A catering company charges RM12.50 per meal. Orders of 50-99 meals cost RM11.50 each, 100-199 cost RM10.50 each, and 200 or more cost RM9.50 each.

Write a complete C++ program that reads the customer name and quantity, validates the quantity, calculates the total, displays a receipt, and repeats until the user chooses to stop. At the end, display the total quantity and total sales.

(15 marks)

END OF QUESTION PAPER